

CPEG/ELEG498/499

Senior Design I and II

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UNIVERSITY OF DELAWARE
ENGINEERING

The degrees of Electrical Engineering and Computer Engineering require that students be prepared for engineering practice through a curriculum culminating in a major design experience. In Senior Design, students apply their accumulated classroom knowledge as well as new state-of-the-art techniques such as machine learning, computer vision, software designed radios, etc., to an Electrical or Computer Engineering design problem. Problems typically involving the design of hardware and/or software systems in many potential domains including: computers/devices, robotics, signal processing, control, and communications.



In Memoriam – Dr. Chase Cotton

Engineering professor remembered for his unique research perspective, student advocacy and mentoring



Chase Cotton, a knowledgeable and passionate researcher, educator, and colleague in the University of Delaware College of Engineering's Department of Electrical and Computer Engineering, passed away on Tuesday, March 14, 2023.



- Canvas page will be published soon!
- Two very important decisions!
 - Team (3 to 4 students per team)
 - Project – should be challenging and something you are passionate about



What makes a Good Project?

- Challenging (year-long effort)
 - Learn something new/stretch yourself
 - Something you can put on your resume
- Able to be broken up and worked on by multiple team members
- Something tangible that you can show off
 - End-of-semester presentations
 - Research Day (\$s!)

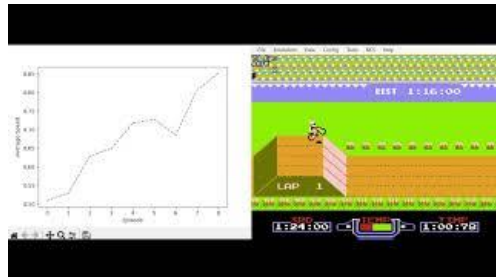
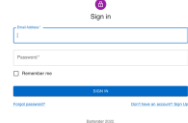
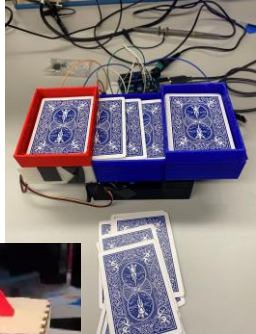
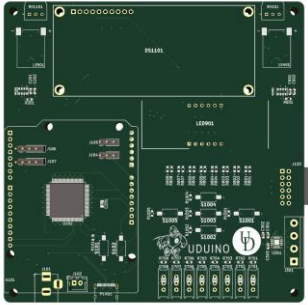


What makes a Good Team

- Complimentary skills
- Shared passion for the project
- Every member communicates with other members of the team
 - All members should show up for meetings with course mentors
 - Each person responsible for their Discord channel



Fall22 - Spring23 Projects



UDUINO

MARS

DECKDEALER

JAKS

TUM

BARTENDER

CRIMEMAP

PACKETSNIFFERS

PLAYAI

SKIMMERSLAYERS

SUPERBLUEHENBROS

VOXT

MEDBASE*

BASYS MX3 replacement

low-noise motor position sensor

card dealer

guitar amplifier with digital effects

robot assistant

Gig marketplace

personal security map

remote network sensor and applications suite

AI video game player

detection of credit card

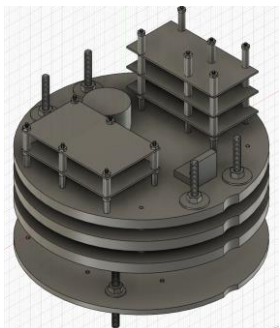
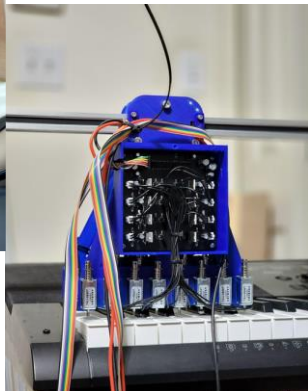
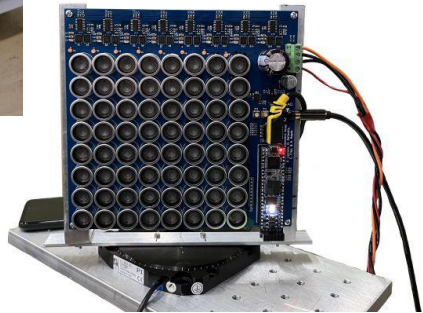
AI legacy game player

sign language recognition

web based health care application framework



Fall21 - Spring22 Projects



FIRE FIGHTERS

M2BAND

PERSONAL PRINTING

STRAGICIANS

uBASS

VITACHAIR

WALL-E

DRIVER ASSIST

DISINFECTING ROBOT

MALWARE ANALYSIS

PANISTS

POTENTIAL DROP

RAGE FROM HOME

wearable device for first responders

fitness band for autistic children

novel 3D printer

smart chessboard

phased array

wireless ECG/pulse ox reporting

personal desktop robot

ODB2 plug-in

automatic disinfection robot using deep UV-C

alternative to hash base tools

device capable of playing a keyboard

NASA payload

VR simulator

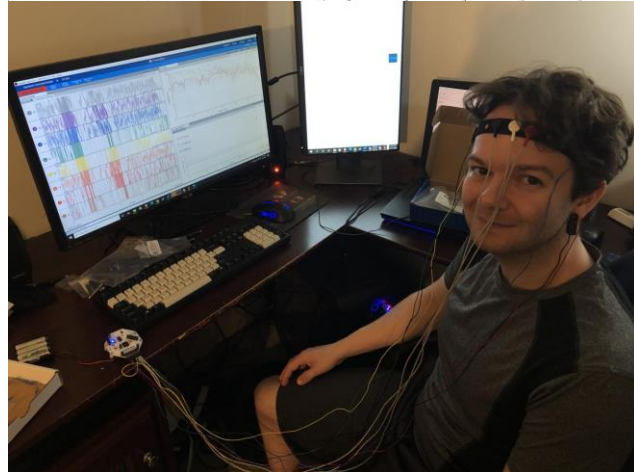


Recent Senior Design Projects

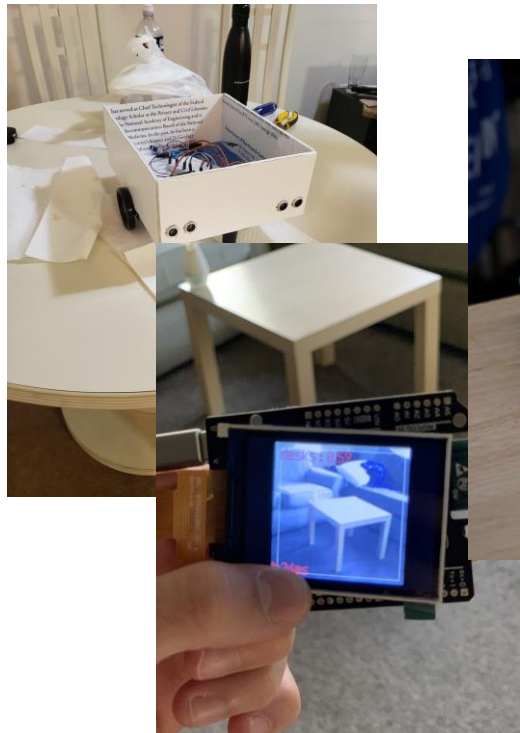
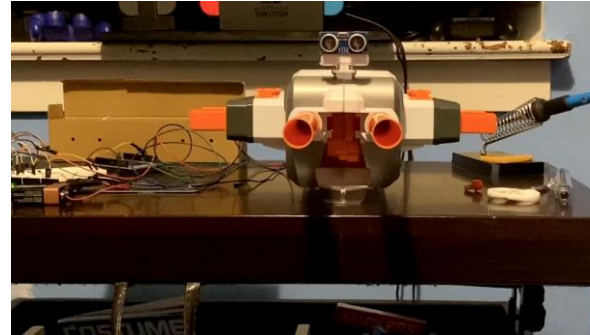
Terminator
Auto Target



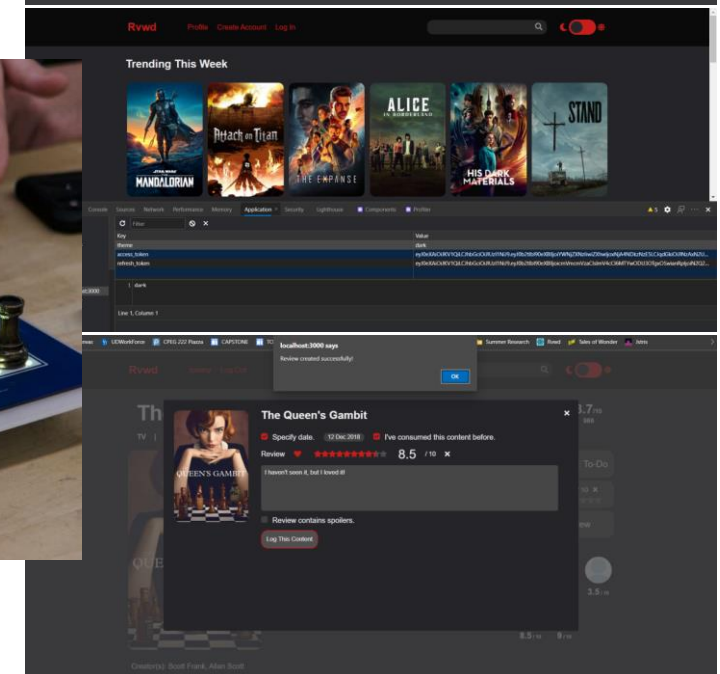
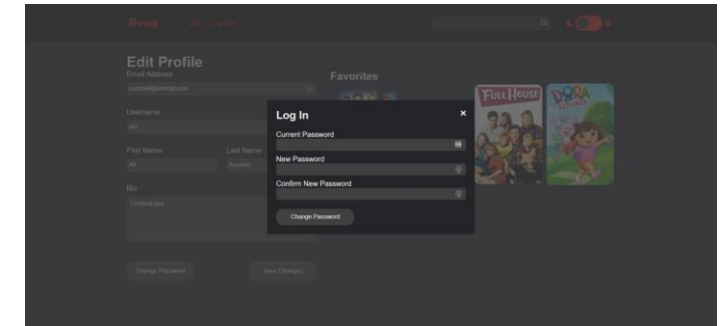
SleepSure



Covid Killer Robot



MediaReviewd



Chess Mate





Recent Senior Design Projects

OUT-OF-THIS-WORLD EXPERIENCE

Article by Julie Stewart | Photo by Evan Krape | Video by Jeffrey Chase and Evan Krape | October 19, 2018

Engineers send senior project to space on research rocket

Just three weeks after earning honors bachelor's degrees from the University of Delaware, four engineers experienced another milestone. They watched in awe as a device they had worked on for nearly a year shot into space on a sounding rocket from Wallops Island, Virginia.

"It was a surreal feeling," said Ryan Beneck, who graduated with a bachelor's degree in electrical engineering. "I kept having to remind myself that something we built was actually inside there."

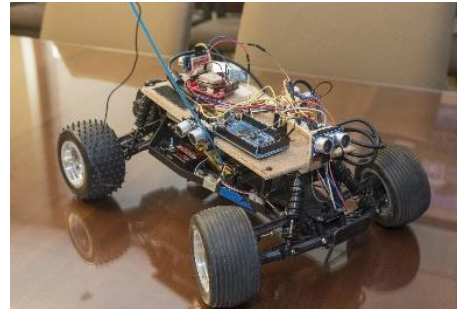
The team constructed the device as part of Senior Design – a capstone course that requires senior electrical and computer engineering students to solve real-world problems and function like small companies.

According to NASA, sounding rockets take their name from the nautical term "to sound," which means to take measurements. Since 1959, NASA-

sponsored space and earth science research has used sounding rockets to test instruments used on satellites and spacecraft and to provide information about the sun, stars, galaxies and Earth's atmosphere and radiation.



The rocket carry the UD engineering team's project launched from NASA's Wallops Flight Facility.



Self driving GPS car



Rock crawling mine detector for the Army



Personal airbag sensor



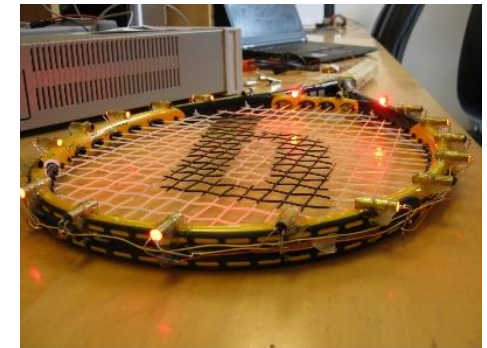
Chemical gas sensor



Laser tag shirt



Gunshot direction sensor



Anti-tennis elbow trainer



Potential F23/S24 Projects

1. Image Processing and Machine Learning (Army ASPIRE program) TPOC: Dr. Justin Wright
2. Autonomous Resupply Caravan (Army ASPIRE program) TPOC: Dr. Sarah Jensen
3. Operations in Degraded Visual Environments (DVE) using Automotive Radars (Army ASPIRE program) TPOC: Dr. Sarah Jensen
4. Cottage Sensor Suite, TPOC: Dr. Mark Bendett
5. SMURF (Sorghum & Maize Under Rotational Force) Sensor. TPOC: Jon Reneau/Dr. Erin Spark's lab
6. Sudden Infant Death Syndrome detector, TPOC: Dr. Martha Hall/Health Design Lab
7. Remote Control Vehicle with Draper tube Chem Sensor (ARMY/ECBC)



The 2024 Army Strategic Program for Innovation, Research & Employment (ASPIRE) is now starting. This year we have all the universities that were involved in 2023 plus we have new entrants UDEL and TTU (Welcome). So this year, as of now, we have 10 university ASPIRE consortium.

GMU, CNU, ODU, CUNY, NCAT, VT, VSU, UDEL, TTU, VCU

ASPIRE, is an educational Capstone series to foster university student (senior undergraduates) development, critical thinking, team building, cross-disciplinary & cross-university projects and Army application skills to develop future workforce for the Army C5ISR needs in Research, Technology & Innovation. **This program requires US Citizenship for students to participate.**

ARMY STRATEGIC PROGRAM FOR INNOVATION, RESEARCH & EMPLOYMENT



ÂSPIRE is a Capstone program to foster senior undergraduate student development, critical thinking, team building, and hands-on training on Army-relevant problems.

The objective of the ÂSPIRE program is to execute science and technology capstone projects relevant to Army C5ISR applications.

Prepare students to enter the workforce in government labs and industry, get recruited as well-trained students to graduate school, and learn about research & technology in areas of interest to C5ISR Center RTI.

ÂSPIRE projects facilitates interaction between the faculty, students, RTI staff, and ÂSPIRE community researchers and help guide educational and research efforts toward innovation.





Image processing and machine learning

Short description of the project/problem:

US Army Soldiers often operate in locations where they are surrounded by textual information in foreign languages, such as on street and shop signs. Many military units have human translators serving or embedded in them to help process this information, however these assets are limited in availability and throughput. At the same time, the US Army is fielding sensors and image processing hardware on most of its platforms. This provides the opportunity to field a real-time, automated foreign text translator that would be available to any Soldier at any time. Under this research topic, students will engineer and demonstrate such a capability using appropriate hardware, software and signal processing algorithms.



Autonomous Resupply Caravan

Short description of the project/problem:

Autonomous vehicles have driven radar technology development and lowered costs. A multitude of cheap automotive radars are now commercially available. One area of interest to be explored is an autonomous vehicle resupply caravan. The project will develop autonomous vehicles capable of following a single leader vehicle using radar guidance. The vehicle caravan should be capable of moving in all direction both forward, sideways, and backwards. Due to the abundance of signals in a modern world, performers would need to be able to identify their waveform from all others. There is interest in camouflaging your signal as well. One approach to this problem would be using machine learning, artificial intelligence, deep learning, and cognitive neural nets.



Operations in Degraded Visual Environments (DVE) using Automotive Radars

Short description of the project/problem:

Degraded Visual Environments (DVE) present significant challenges due to lack of situational awareness. Such environments include low light, fog, dust, rain, snow, and other visual obscurants. There is a need to improve situational awareness for automobile usage and helicopter landings. Imaging in the millimeter wave region of the spectrum has demonstrated great utility for the ability to see through such obscurants with low attenuation, thereby providing navigational and operational cues. Nonetheless, current millimeter wave imaging technology remains complex, expensive, and high size, weight and power (SWaP); significant reductions in these parameters are required to facilitate greater deployment opportunities and open new applications and markets. To this end, increased leverage of commercial off-the-shelf (COTS) components is needed. The recent proliferation of collision avoidance radar systems within the automotive industry has created a huge market for millimeter wave components, bringing production to scale and driving down prices. This project will explore using automotive radars for operations in DVE. The project deliverables will include a prototype demonstration capable of identifying metal target through DVE.



Cottage / Apartment Sensor Suite for Health Monitoring

The problem being addressed with this initiative is the challenge of enabling independent living for a growing aging population and appropriate avoidance of institutionalization. Aging brings physiological, social and psychological changes that frequently result in patients/residents being unsafe in their own homes. With the use of technology, it should be possible for clinicians and/or caregivers to be in closer contact with their patient/resident and feel confident that any significant changes in the condition of or risks to their charge will be promptly identified so that appropriate action can be initiated. Enabling this capability will allow more aging individuals to safely remain in their home and avoid institutionalization.



- Canvas page will be published soon!
- Two very important decisions!
 - Team (3 to 4 students per team)
 - Project
- Discord Channels will be used to Log weekly progress
- Project Proposal due Monday 9/18



Tentative Class Grading

- 15% PROPOSAL – due September 18
- 20% The individual student's discord documentation of week-by-week activities
- 20% FINAL PRESENTATION (15 minutes + 5 minutes of questions) - December 5th and 7th
- 30% FINAL REPORT/RESULTS - due December 15th
- 15% WORKING PROTOTYPE OR PARTIAL PROTOTYPE/DESIGN FILES (code, 3D designs, PCB designs, etc.) - due December 15th

INDIVIDUALS MAY LOOSE ONE LETTER GRADE IF THEY FAIL TO PARTICIPATE, CONTRIBUTE, AND DOCUMENT ACTIVITIES TOWARD THEIR GROUP'S PROJECT (e.g. A to B, B to C, etc.)



- 8/31 – Dr. Maruca (PHYS/NASA) /Dr. Bendett (Senior Living Sensor Suite)
- 9/5 – Jon Reneau (Sparks Lab, Agri Sensor) /Dr. Jensen (ASPIRE Army Projects)
- 9/14 – Fall Career & Internship Fair, 3:00-6:00 PM
- 9/19 – Civil, Environmental & Construction Engineering Career Fair, 4:30-6:30 PM



UNIVERSITY OF DELAWARE
ENGINEERING

Congratulations and Hang in There!





- Questions?
- Team Forming/Project Ideation